

CMU 10-703: Deep Reinforcement Learning

Lecture 8: PPO and Off-Policy Actor-Critic

Max Simchowitz

Carnegie Mellon University

Lecture Map



Policy proximity

KL penalties and ratio clipping control how far the policy moves.

Directional clipping

The sign of the advantage determines which ratio boundary is active.

Data reuse

Replay data supports critic learning for a target policy.

1.

Performance Difference and KL Direction

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Performance Difference Lemma

$$\begin{aligned} J(\pi) - J(\mu) &= \mathbb{E}_{\tau \sim P_\pi} \left[\sum_{t=0}^{\infty} \gamma^t A^\mu(s_t, a_t) \right] \\ &= \frac{1}{1 - \gamma} \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi(\cdot|s)}} [A^\mu(s, a)]. \end{aligned}$$

Advantage

A^μ evaluates an action relative to the data-collection policy.

Occupancy

d^π weights states visited by the target policy.

Two KL Directions

AWR: $D_{\text{KL}}(\pi \parallel \mu)$

$$\max_{\pi} \mathbb{E}_{s \sim d^{\mu}} [\mathbb{E}_{a \sim \pi} [A^{\mu}] - \alpha D_{\text{KL}}(\pi \parallel \mu)]$$

closed-form policy update

PPO/TRPO: $D_{\text{KL}}(\mu \parallel \pi)$

$$\max_{\theta} \mathbb{E}_{s \sim d^{\mu}} [\mathbb{E}_{a \sim \pi_{\theta}} [A^{\mu}] - \alpha D_{\text{KL}}(\mu \parallel \pi_{\theta})]$$

gradient-based policy update

KL direction changes both the optimization problem and the algorithm.

Advantage-Weighted Regression

Closed-form optimum

$$\pi^*(a \mid s) \propto \mu(a \mid s) \exp\left(\frac{A^\mu(s, a)}{\alpha}\right).$$

behavior policy
 $\mu(a \mid s)$

$\times$

advantage weight
 $\exp(A^\mu/\alpha)$

$\longrightarrow$

new policy
 π^*

Reverse KL removes this closed form and leads to the PPO/TRPO optimization path.

2.

Surrogate Policy Optimization

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Policy Gradient Theorem

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) A^{\pi_{\theta}}(s_t, a_t) \right].$$

The expectation requires trajectories from the current policy.

Policy Gradient Theorem

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) A^{\pi_{\theta}}(s_t, a_t) \right].$$

Score term

$\nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t)$ changes action probability.

Weight

$A^{\pi_{\theta}}(s_t, a_t)$ determines the update direction.

PPO approximates this update with data collected by an earlier policy.

Action Importance Sampling

$$\mathbb{E}_{a \sim \pi_\theta}[f(a)] = \mathbb{E}_{a \sim \mu} \left[\frac{\pi_\theta(a \mid s)}{\mu(a \mid s)} f(a) \right].$$

Change the action distribution by reweighting samples from μ .

Action Importance Sampling

$$\mathbb{E}_{a \sim \pi_\theta}[f(a)] = \mathbb{E}_{a \sim \mu} \left[\frac{\pi_\theta(a \mid s)}{\mu(a \mid s)} f(a) \right].$$

Importance ratio. $r_t(\theta) = \pi_\theta(a_t \mid s_t) / \mu(a_t \mid s_t)$.

The ratio records how the new policy revalues each sampled action.

Action Importance Sampling

$$\mathbb{E}_{a \sim \pi_\theta}[f(a)] = \mathbb{E}_{a \sim \mu} \left[\frac{\pi_\theta(a \mid s)}{\mu(a \mid s)} f(a) \right].$$

Importance ratio. $r_t(\theta) = \pi_\theta(a_t \mid s_t) / \mu(a_t \mid s_t)$.

$$r_t(\theta) = 1 \quad \Longleftrightarrow \quad \pi_\theta(a_t \mid s_t) = \mu(a_t \mid s_t).$$

A ratio near one indicates a local policy update.

Conservative Policy Iteration Surrogate

$$L^{\text{CPI}}(\theta) = \mathbb{E}_{(s,a) \sim \mu} \left[\frac{\pi_{\theta}(a \mid s)}{\mu(a \mid s)} A^{\mu}(s, a) \right].$$



The surrogate reuses old actions while treating old-policy advantages as fixed labels.

State Distribution Approximation

Exact change of measure

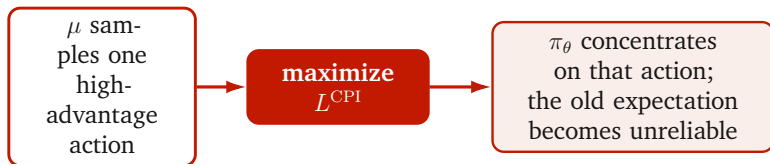
$$\mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}}[A^\mu] = \mathbb{E}_{\substack{s \sim d^\mu \\ a \sim \mu}} \left[\frac{d^\pi(s)}{d^\mu(s)} \frac{\pi(a | s)}{\mu(a | s)} A^\mu(s, a) \right].$$

PPO approximation

$$d^\pi \approx d^\mu, \quad \text{retain only } \frac{\pi(a | s)}{\mu(a | s)}.$$

State-density ratios are high-dimensional and can accumulate severe variance over long horizons.

Unconstrained Surrogate Failure



A large policy change invalidates the distributional approximation used to define the surrogate.

Policy proximity keeps the importance correction informative.

Trust Region Policy Optimization

$$\begin{aligned} \max_{\theta} \quad & \mathbb{E}_{(s,a) \sim \mu} \left[\frac{\pi_{\theta}(a \mid s)}{\mu(a \mid s)} A^{\mu}(s, a) \right] \\ \text{subject to} \quad & \mathbb{E}_{s \sim \mu} [D_{\text{KL}}(\mu(\cdot \mid s) \parallel \pi_{\theta}(\cdot \mid s))] \leq \delta. \end{aligned}$$

Constraint

Explicit average KL trust region

Approximation

Second-order local KL model

Solver

Conjugate-gradient direction

Source: Schulman et al., "Trust Region Policy Optimization," ICML 2015.

TRPO Optimization Cost



Benefit

The explicit trust region supports monotonic policy-improvement guarantees under the approximation.

Cost

Each update needs curvature-vector products and a constrained optimization routine.

PPO replaces the second-order trust-region solve with a first-order clipped objective.

3.

Proximal Policy Optimization

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Basic Ratio Clipping

$$L_{\text{basic}}^{\text{CLIP}}(\theta) = \mathbb{E}_{(s,a) \sim \mu} [\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A^\mu(s, a)].$$



$\epsilon \approx 0.1\text{--}0.2$ sets the allowed ratio change.

Source: Schulman et al., "Proximal Policy Optimization Algorithms," 2017.

Failure of Basic Clipping

Positive-advantage sample below the lower bound

$$A_t^\mu > 0, \quad r_t(\theta) < 1 - \epsilon.$$

$$\text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)A_t^\mu = (1 - \epsilon)A_t^\mu.$$

Desired update

Increase $\pi_\theta(a_t \mid s_t)$ because $A_t^\mu > 0$.

Basic clipping

The objective is constant in θ , so its gradient is zero.

PPO Clipped Surrogate Objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t [\min(r_t(\theta)A_t^\mu, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t^\mu)] .$$

Compare the ordinary surrogate with a clipped surrogate sample by sample.

PPO Clipped Surrogate Objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t [\min(r_t(\theta) A_t^\mu, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t^\mu)] .$$

The minimum clips only a policy change that moves too far in the direction favored by the advantage sign.

The unfavorable branch remains active, allowing a corrective gradient back toward the trust region.

Positive-Advantage Clipping

$$A_t^\mu > 0 \implies \min(r_t A_t^\mu, \text{clip}(r_t) A_t^\mu).$$

$r_t < 1 - \epsilon$

$1 - \epsilon \leq r_t \leq 1 + \epsilon$

$r_t > 1 + \epsilon$

Unclipped branch.

Increase the probability.

A good action that became underweighted receives a corrective gradient.

Positive-Advantage Clipping

$$A_t^\mu > 0 \implies \min(r_t A_t^\mu, \text{clip}(r_t) A_t^\mu).$$

$$r_t < 1 - \epsilon$$

Unclipped branch.

Increase the probability.

$$1 - \epsilon \leq r_t \leq 1 + \epsilon$$

Unclipped branch.

Increase the probability.

$$r_t > 1 + \epsilon$$

Inside the trust region, the ordinary policy-gradient direction remains active.

Positive-Advantage Clipping

$$A_t^\mu > 0 \implies \min(r_t A_t^\mu, \text{clip}(r_t) A_t^\mu).$$

$$r_t < 1 - \epsilon$$

Unclipped branch.

Increase the probability.

$$1 - \epsilon \leq r_t \leq 1 + \epsilon$$

Unclipped branch.

Increase the probability.

$$r_t > 1 + \epsilon$$

Clipped plateau. Stop

increasing the probability.

The upper cap limits further probability increases for a good action.

Negative-Advantage Clipping

$$A_t^\mu < 0 \implies \min(r_t A_t^\mu, \text{clip}(r_t) A_t^\mu).$$

$r_t < 1 - \epsilon$

$1 - \epsilon \leq r_t \leq 1 + \epsilon$

$r_t > 1 + \epsilon$

Clipped plateau. Stop
decreasing the probability.

The lower floor limits further probability decreases for a bad action.

Negative-Advantage Clipping

$$A_t^\mu < 0 \implies \min(r_t A_t^\mu, \text{clip}(r_t) A_t^\mu).$$

$$r_t < 1 - \epsilon$$

Clipped plateau. Stop
decreasing the probability.

$$1 - \epsilon \leq r_t \leq 1 + \epsilon$$

Unclicked branch.
Decrease the probability.

$$r_t > 1 + \epsilon$$

Inside the trust region, the ordinary policy-gradient direction remains active.

Negative-Advantage Clipping

$$A_t^\mu < 0 \implies \min(r_t A_t^\mu, \text{clip}(r_t) A_t^\mu).$$

$$r_t < 1 - \epsilon$$

Clipped plateau. Stop
decreasing the probability.

$$1 - \epsilon \leq r_t \leq 1 + \epsilon$$

Unclipped branch.
Decrease the probability.

$$r_t > 1 + \epsilon$$

Unclipped branch. Pull the
probability down.

An overweighted bad action receives a corrective gradient.

Clipping Behavior by Region

A_t^μ	Ratio region	Active branch	Update effect
> 0	$r_t > 1 + \epsilon$	clipped	zero: upper cap
> 0	inside interval	unclipped	increase probability
> 0	$r_t < 1 - \epsilon$	unclipped	pull probability up
< 0	$r_t < 1 - \epsilon$	clipped	zero: lower floor
< 0	inside interval	unclipped	decrease probability
< 0	$r_t > 1 + \epsilon$	unclipped	pull probability down

Clip a positive-advantage sample only above the interval; clip a negative-advantage sample only below it.

Directional Clipping Rationale

Good action below the floor

$$A_t^\mu > 0, \quad r_t < 1 - \epsilon$$

Because μ assigned this action substantial probability, its advantage estimate is comparatively well supported. PPO allows a corrective increase.

Bad action below the floor

$$A_t^\mu < 0, \quad r_t < 1 - \epsilon$$

Further decreasing one action gives little control over where its probability mass moves. PPO stops the decrease.

The symmetric ratio interval produces asymmetric gradients because the advantage changes which branch is pessimistic.

KL Weight and Clip Radius

Penalized reverse KL

$$\text{return surrogate} - \alpha D_{\text{KL}}(\mu \parallel \pi_{\theta})$$

Larger α penalizes policy movement more strongly.

Clipped ratio

$$r_t(\theta) \in [1 - \epsilon, 1 + \epsilon]$$

Larger ϵ permits a wider local probability change.

A larger clip radius plays the same conceptual role as a weaker KL penalty.

Policy Rollouts

Inputs. $\pi_{\theta_0}, V_{\phi_0}, \epsilon, \alpha_{\theta}, \alpha_{\phi}, T, K, M.$

freeze θ_{old}

Each PPO iteration begins from a fixed behavior policy.

Policy Rollouts

Inputs. $\pi_{\theta_0}, V_{\phi_0}, \epsilon, \alpha_\theta, \alpha_\phi, T, K, M.$

freeze θ_{old}

collect T transitions

$$\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}_{t=0}^{T-1}, \quad a_t \sim \pi_{\theta_{\text{old}}}(\cdot \mid s_t).$$

The fixed policy defines the rollout distribution and denominator of the ratio

Policy Rollouts

Inputs. $\pi_{\theta_0}, V_{\phi_0}, \epsilon, \alpha_{\theta}, \alpha_{\phi}, T, K, M.$

freeze θ_{old}

collect T transitions

compute returns R_t

$$\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}_{t=0}^{T-1}, \quad a_t \sim \pi_{\theta_{\text{old}}}(\cdot \mid s_t).$$

Returns provide targets for the value-function update.

Policy Rollouts

Inputs. $\pi_{\theta_0}, V_{\phi_0}, \epsilon, \alpha_{\theta}, \alpha_{\phi}, T, K, M.$

freeze θ_{old}

collect T transitions

compute returns R_t

estimate A_t , e.g. GAE

$$\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}_{t=0}^{T-1}, \quad a_t \sim \pi_{\theta_{\text{old}}}(\cdot \mid s_t).$$

Advantage estimates provide weights for the policy update.

PPO Mini-Batch Updates

1. Repeat for K epochs over mini-batches $\mathcal{B}$ with $|\mathcal{B}| = M$.
2. Compute $r_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$.

PPO Mini-Batch Updates

1. Repeat for K epochs over mini-batches $\mathcal{B}$ with $|\mathcal{B}| = M$.
2. Compute $r_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$.
3. Set $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta L^{\text{CLIP}}$.

$$\frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} \min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t) .$$

PPO Mini-Batch Updates

1. Repeat for K epochs over mini-batches $\mathcal{B}$ with $|\mathcal{B}| = M$.
2. Compute $r_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$.
3. Set $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta L^{\text{CLIP}}$.
4. Update ϕ with learning rate α_ϕ on the return MSE.

$$\frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} \min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t) .$$

PPO Mini-Batch Updates

1. Repeat for K epochs over mini-batches $\mathcal{B}$ with $|\mathcal{B}| = M$.
2. Compute $r_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$.
3. Set $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta L^{\text{CLIP}}$.
4. Update ϕ with learning rate α_ϕ on the return MSE.
5. Set $\theta_{\text{old}} \leftarrow \theta$ before the next rollout.

$$\frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} \min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t) .$$

PPO with Adaptive KL Penalty

$$L^{\text{CLIP+KL}}(\theta) = \mathbb{E}_t [\min(r_t A_t, \text{clip}(r_t) A_t)] \\ - \beta \mathbb{E}_t [D_{\text{KL}}(\mu(\cdot \mid s_t) \parallel \pi_\theta(\cdot \mid s_t))] .$$

Observed KL too large

Increase β when $D_{\text{KL}} > \delta_{\text{target}}$.

Observed KL too small

Decrease β when $D_{\text{KL}} < \delta_{\text{target}}/1.5$.

The clip-only variant performed comparably or better in the original PPO study and became the default.

4.

DAPO and Asymmetric Clipping

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Directional Effects of Symmetric Clipping

$$\text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)$$

Positive advantage

The upper bound limits probability increases.

Negative advantage

The lower bound limits probability decreases.

The interval is symmetric around one, but the minimum objective activates only one side for each advantage sign.

DAPO Asymmetric Bounds

$$\text{clip}(r_{\theta}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}), \quad \epsilon_{\text{high}} > \epsilon_{\text{low}}.$$

Looser upper cap

$$\epsilon_{\text{high}} = 0.28$$

Allow larger probability increases for positive-advantage actions.

Source: Yu et al., "DAPO: An Open-Source LLM Reinforcement Learning System at Scale," 2025.

Tighter lower floor

$$\epsilon_{\text{low}} = 0.20$$

Limit aggressive probability decreases for negative-advantage actions.

DAPO in Large Action Spaces



Trajectory scale

More sampled trajectories can improve advantage estimates for uncommon actions.

Entropy collapse

DAPO also uses entropy regularization to preserve policy diversity.

Asymmetric clipping allocates more update range to promoting good actions than suppressing bad ones.

5.

Off-Policy Actor-Critic

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Two Forms of the Performance Difference Lemma

Form 1: PPO and AWR

$$J(\pi) - J(\mu) = \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [A^\mu(s, a)].$$

Form 1 evaluates target-policy states with the behavior policy's advantage.

Two Forms of the Performance Difference Lemma

Form 1: PPO and AWR

$$J(\pi) - J(\mu) = \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [A^\mu(s, a)].$$

Form 2: off-policy actor-critic

$$J(\pi) - J(\mu) = \mathbb{E}_{\substack{s \sim d^\mu \\ a \sim \pi}} [A^\pi(s, a)].$$

Form 2 evaluates behavior-policy states with the target policy's advantage.

Data and Advantage Roles

Form 1

$$s \sim d^{\pi}, \quad A^{\mu}(s, a)$$

The target policy determines state occupancy; the data policy supplies the advantage estimate.

Form 2

$$s \sim d^{\mu}, \quad A^{\pi}(s, a)$$

The replay distribution supplies states; a critic must evaluate the target policy.

Form 2 exchanges occupancy correction for the problem of learning Q^{π} from off-policy data.

Off-Policy Actor-Critic Architecture



Store reusable transitions instead of discarding each rollout.

Off-Policy Actor-Critic Architecture



The critic estimates target-policy values from transitions collected by the behavior policy.

Bellman bootstrapping trains Q_ϕ toward Q^π .

Off-Policy Actor-Critic Architecture



The critic estimates target-policy values from transitions collected by the behavior policy.

The actor chooses actions that the learned critic values highly.

Critic Bellman Equation

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{\substack{s' \sim P(\cdot|s, a) \\ a' \sim \pi(\cdot|s')}} [Q^\pi(s', a')].$$



The target action comes from π_θ , even when the transition came from μ .

Temporal-Difference Critic Loss

$$y = r + \gamma \mathbb{E}_{a' \sim \pi_{\theta}(\cdot | s')} [Q_{\phi^-}(s', a')].$$

$$\mathcal{L}(\phi) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(Q_{\phi}(s, a) - y)^2 \right].$$

Target network

Q_{ϕ^-} changes slowly to stabilize bootstrapping.

Relation to DQN

Sample or average $a' \sim \pi_{\theta}$ instead of taking an $\arg \max$.

Actor Learning from the Critic

Critic objective for the actor

$$\max_{\theta} \mathbb{E}_{\substack{s \sim d^{\mu} \\ a \sim \pi_{\theta}(\cdot|s)}} [Q_{\phi}(s, a)].$$

Equivalent advantage-centered view

$$A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s), \quad V^{\pi}(s) = \mathbb{E}_{a \sim \pi} [Q^{\pi}(s, a)].$$

The actor increases probability on actions that the critic rates above the policy's state value.

TD3 and SAC



TD3

Deterministic actor with critic techniques that reduce overestimation and target error.

SAC

Stochastic actor with entropy regularization and off-policy critic learning.

Lecture 9 develops these two realizations of the form-2 actor-critic framework.

Summary

Proximal updates

- TRPO constrains reverse KL explicitly.
- PPO uses a first-order clipped surrogate.
- The advantage sign selects the active boundary.

DAPO

- $\epsilon_{\text{high}} > \epsilon_{\text{low}}$.
- Positive updates receive more range.
- Entropy regularization counters collapse.

Off-policy actor-critic

- Replay supplies states from d^μ .
- The critic estimates Q^π .
- TD3 and SAC implement the form-2 bridge.

The two PDL forms connect proximal on-policy updates to replay-based actor-critic methods.